

CHARACTER-LEVEL DIAGNOSTIC

Restoration Impact on HTR Error Patterns

70% Total Error Reduction

19,417 → 5,824 errors

88.6%

Deletion Recovery

97.8%

Characters Visible

DEGRADED

Restoration

RESTORED

92.8% Deletion

35.2% Deletion

7.2% Substitution

64.8% Substitution

KEY INSIGHTS

Restoration makes invisible text visible (76.4% → 2.2% missing)

Deletion errors are catastrophic (text lost)

Substitution errors are recoverable (text visible but confused)

Net benefit: 70% error reduction despite substitution increase

All word positions benefit (middle 71.2%, ends 68.2%, start 57.8%) | $p < 0.001$